

Jacob Scott Clark

Senior Full Stack Engineer

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SUMMARY

Senior Full Stack & AI Engineer with 8 years of experience delivering 15+ scalable web and cloud applications in Node.js, NestJS, Python, React, TypeScript, MongoDB and PostgreSQL. Proficient in C++/Python for AI, computer vision and LLMs, including YOLO/OpenCV, ONNX/TensorRT/libtorch, AWS, Docker and model fine-tuning. Proven impact: 3× faster processing, +35% serving speed, 40% fewer errors, 30% higher accuracy, and 47% faster access.

WORK EXPERIENCE

GRUPO LEO | Panama, Panama

Senior AI Full Stack Engineer

Oct 2022 - Present

- Designed relational and document schemas (PostgreSQL/MySQL, MongoDB); wrote migration plans and query optimizations to keep p95 latencies low.
- Developed AI-powered applications with Node.js, NestJS, React, TypeScript, PostgreSQL, MongoDB and LangChain, using Kafka for event streaming and improving data validation processing speed by 30%.
- Implemented REST services (Flask) for authentication, billing hooks, and streaming endpoints; added pagination, caching (Redis), and request throttling for scale.
- Productized LLM and CV workloads, fine-tuning for tool use, prompt/rubric design, safe-guardrails, packaged models for real-time inference.
- Forged an automated provisioning solution via Terraform, Docker, and GitOps, streamlining CI/CD pipelines and shrinking deployment times by 40%.

Evoload | Bucharest, Romania

Senior Full Stack Engineer

Apr 2021 - Oct 2022

- Prepared secure web apps and admin consoles using React, TypeScript, and TailwindCSS with enhanced UI/UX for real-time AI controls, analytics and role-based access, showcasing front-end development skills..
- Orchestrated the launch and monitoring of machine learning models using Python, TensorFlow, and PyTorch, elevating model performance accuracy by 25% in production environments.
- Engineered scalable microservices using NestJS, React, and MongoDB on AWS, processing 20K+ REST APIs requests monthly and improving system reliability.
- Streamlined CI/CD pipelines with Terraform and GitHub Actions, cutting deployment time by 30%.
- Combined Elasticsearch for enhanced search capabilities, improving search query response time by 75%.
- Produced efficient SQL and NoSQL schemas for microservices, increasing data modeling and retrieval speed by 80%.
- Upgraded real-time communication features using WebSockets, reducing latency by 65% and improving user experience.

Computer Audio Inc | NC, United State

Full Stack Engineer

Jan 2014 - Apr 2021

- Launched full-stack web applications using React and Django REST Framework, improving scalability and response time by 35%.

- Constructed REST and GraphQL APIs with MongoDB and Redis for background jobs, reducing latency by 30% and improving caching strategies.
- Administered Kubernetes clusters through Rancher, maintaining 95% uptime across all environments, including Linux System Administration and Networking.
- Designed CI/CD pipelines with GitHub Actions and Docker, accelerating releases by 45%, leveraging GitHub for version control and collaboration, and utilizing Webpack for bundling.
- Improved UI/UX design of web applications, resulting in 80% higher user satisfaction and engagement.

Dowd Technology Group | NC, United State

Software Engineer

Apr 2012 - Jan 2014

- Deployed and optimized React-based frontend integrated with REST APIs, enhancing API performance and achieving 21% higher engagement.
- Developed and maintained a customer database using Microsoft SQL Server and PostgreSQL, improving data accessibility for sales teams by 75%.
- Improved API latency by 25% through caching, Redis optimization, and performance tuning.
- Architected and built multi-tenant distributed systems with strict data isolation, ensuring compliance with data privacy regulations, following Architectural principles.
- Standardized UI components with React libraries, implementing robust testing practices and reducing frontend development time by 20%.
- Collaborated cross-functionally to integrate and optimize APIs and SDKs, including Salesforce APIs and Nest.js microservices, boosting delivery speed by 30%.

SELECTED ACHIEVEMENTS

- Established and maintained AI full-stack SaaS applications using the MERN stack (Express, NestJS, Python, React, Next.js, PostgreSQL, MongoDB), using ClickHouse for analytics and enhancing processing speed by 35%.
- Architected Python-based backend and AI automation services integrating LLMs, LangChain, and TensorFlow to streamline data analysis workflows.
- Delivered cloud-native solutions on AWS, Azure, and Google Cloud Platform (GCP) leveraging microservices and serverless architectures, achieving 99.9% uptime and 40% cost optimization.

CORE SKILLS

Languages: C++, C#, Python, PHP, Typescript, JavaScript (ES6+), Bash

Frontend: React.js, Next.js, Vue.js, Tailwind CSS, Sass, Bootstrap, Chakra UI, Storybook, Three.js, WebGL

State Management: Redux, Zustand, React Query, MobX, TanStack Query

Backend: Node.js, NestJS, Django, Flask, FastAPI, REST APIs, WebSockets, Microservices, TypeORM

AI: TensorFlow, PyTorch, Scikit-learn, NLTK, LLMs, LangChain, MLflow, Kubeflow, OpenCV, NumPy

DevOps: AWS (S3, EC2, Lambda), Azure Cloud, Puppet, Ansible, Docker, Kubernetes, Terraform, CI/CD

Testing & Tools: Jest, Mocha, Cypress, React Testing Library, GitHub Actions, Vite, Webpack, ESLint, Husky

Other: SQL, NoSQL, GraphQL, MongoDB, MySQL, PostgreSQL, Redis, Web3.js, Vercel, Vite, Elasticsearch

Collaboration: Git, Gitlab, monorepo, Agile/Scrum, Figma, Jira, Code Reviews, Cross-functional Teams

EDUCATION

University of North Carolina Wilmington

Sep 2006 – Apr 2012

Bachelor's Degree in Computer Science